

Comparativa de Semánticas Formales (NS vs SOS)

1. Semántica Natural (NS)

$$\begin{aligned}
\mathcal{T}_1 &= \langle x := 1, \sigma_0 \rangle \longrightarrow \sigma_0[x \mapsto \mathcal{A}[[1]]\sigma_0] \text{ [ass}_{\text{ns}}] \\
\mathcal{T}_2 &= \langle y := 5, \sigma_1 \rangle \longrightarrow \sigma_1[y \mapsto \mathcal{A}[[5]]\sigma_1] \text{ [ass}_{\text{ns}}] \\
\mathcal{T}_3 &= \frac{\langle \varepsilon, \sigma_3 \rangle \longrightarrow_D \sigma_3 \text{ [none}_{\text{ns}}]}{\langle \mathbf{var} \ y := 2 ;, \sigma_2 \rangle \longrightarrow_D \sigma_2[y \mapsto \mathcal{A}[[2]]\sigma_2] \text{ [var}_{\text{ns}}]} \\
\mathcal{T}_4 &= \langle x := x + 5, \sigma_3 \rangle \longrightarrow \sigma_3[x \mapsto \mathcal{A}[[x + 5]]\sigma_3] \text{ [ass}_{\text{ns}}] \\
\mathcal{T}_5 &= \langle x := 0, \sigma_4 \rangle \longrightarrow \sigma_4[x \mapsto \mathcal{A}[[0]]\sigma_4] \text{ [ass}_{\text{ns}}] \\
\mathcal{T}_6 &= \langle y := y + 1, \sigma_5 \rangle \longrightarrow \sigma_5[y \mapsto \mathcal{A}[[y + 1]]\sigma_5] \text{ [ass}_{\text{ns}}] \\
\mathcal{T}_7 &= \langle x := x + 1, \sigma_6 \rangle \longrightarrow \sigma_6[x \mapsto \mathcal{A}[[x + 1]]\sigma_6] \text{ [ass}_{\text{ns}}] \\
\mathcal{T}_8 &= \langle y := y + 1, \sigma_7 \rangle \longrightarrow \sigma_7[y \mapsto \mathcal{A}[[y + 1]]\sigma_7] \text{ [ass}_{\text{ns}}] \\
\mathcal{T}_9 &= \langle x := x + 1, \sigma_8 \rangle \longrightarrow \sigma_8[x \mapsto \mathcal{A}[[x + 1]]\sigma_8] \text{ [ass}_{\text{ns}}] \\
\mathcal{T}_{10} &= \langle y := y + 1, \sigma_{10} \rangle \longrightarrow \sigma_{10}[y \mapsto \mathcal{A}[[y + 1]]\sigma_{10}] \text{ [ass}_{\text{ns}}] \\
\mathcal{T}_{11} &= \langle \mathbf{for} \ x := x + 1 \ \mathbf{to} \ 2 \ \mathbf{do} \ y := y + 1, \sigma_{11} \rangle \longrightarrow \sigma_{11} \text{ [for - ff}_{\text{ns}}] \\
\mathcal{T}_{12} &= \frac{\mathcal{T}_9 \quad \mathcal{T}_{10} \quad \mathcal{T}_{11}}{\langle \mathbf{for} \ x := x + 1 \ \mathbf{to} \ 2 \ \mathbf{do} \ y := y + 1, \sigma_8 \rangle \longrightarrow \sigma_{11} \text{ [for - tt}_{\text{ns}}]} \\
\mathcal{T}_{13} &= \frac{\mathcal{T}_7 \quad \mathcal{T}_8 \quad \mathcal{T}_{12}}{\langle \mathbf{for} \ x := x + 1 \ \mathbf{to} \ 2 \ \mathbf{do} \ y := y + 1, \sigma_6 \rangle \longrightarrow \sigma_{11} \text{ [for - tt}_{\text{ns}}]} \\
\mathcal{T}_{14} &= \frac{\mathcal{T}_5 \quad \mathcal{T}_6 \quad \mathcal{T}_{13}}{\langle \mathbf{for} \ x := 0 \ \mathbf{to} \ 2 \ \mathbf{do} \ y := y + 1, \sigma_4 \rangle \longrightarrow \sigma_{11} \text{ [for - tt}_{\text{ns}}]} \\
\mathcal{T}_{15} &= \frac{\mathcal{T}_4 \quad \mathcal{T}_{14}}{\langle S_1, \sigma_3 \rangle \longrightarrow \sigma_{11} \text{ [comp}_{\text{ns}}]} \\
\mathcal{T}_{16} &= \frac{\mathcal{T}_3 \quad \mathcal{T}_{15}}{\langle S_2, \sigma_2 \rangle \longrightarrow \sigma_{11}[\text{DV}(\mathbf{var} \ y := 2 ;) \mapsto \sigma_2] \text{ [block}_{\text{ns}}]} \\
\mathcal{T}_{17} &= \frac{\mathcal{T}_2 \quad \mathcal{T}_{16}}{\langle S_3, \sigma_1 \rangle \longrightarrow \sigma_{11} \text{ [comp}_{\text{ns}}]} \\
\mathcal{T}_{18} &= \frac{\mathcal{T}_1 \quad \mathcal{T}_{17}}{\langle S_4, \sigma_0 \rangle \longrightarrow \sigma_{12} \text{ [comp}_{\text{ns}}]}
\end{aligned}$$

Sentencias Abreviadas

$S_1 = (x := x + 5 ; \mathbf{for} \ x := 0 \ \mathbf{to} \ 2 \ \mathbf{do} \ y := y + 1)$
 $S_2 = \mathbf{begin} \ \mathbf{var} \ y := 2 ; (x := x + 5 ; \mathbf{for} \ x := 0 \ \mathbf{to} \ 2 \ \mathbf{do} \ y := y + 1) \ \mathbf{end}$
 $S_3 = (y := 5 ; \mathbf{begin} \ \mathbf{var} \ y := 2 ; (x := x + 5 ; \mathbf{for} \ x := 0 \ \mathbf{to} \ 2 \ \mathbf{do} \ y := y + 1) \ \mathbf{end})$
 $S_4 = (x := 1 ; y := 5 ; \mathbf{begin} \ \mathbf{var} \ y := 2 ; (x := x + 5 ; \mathbf{for} \ x := 0 \ \mathbf{to} \ 2 \ \mathbf{do} \ y := y + 1) \ \mathbf{end}) \quad (\mathbf{S}_{\text{ini}})$

Leyenda de Estados

- $\sigma_0 \equiv \emptyset$ (σ_{ini})
- $\sigma_1 \equiv \sigma_0[x \mapsto \mathcal{A}[\![1]\!]\sigma_0] = \{x \mapsto 1\}$
- $\sigma_2 \equiv \sigma_1[y \mapsto \mathcal{A}[\![5]\!]\sigma_1] = \{x \mapsto 1, y \mapsto 5\}$
- $\sigma_3 \equiv \sigma_2[y \mapsto \mathcal{A}[\![2]\!]\sigma_2] = \{x \mapsto 1, y \mapsto 2\}$
- $\sigma_4 \equiv \sigma_3[x \mapsto \mathcal{A}[\![x + 5]\!]\sigma_3] = \{x \mapsto 6, y \mapsto 2\}$
- $\sigma_5 \equiv \sigma_4[x \mapsto \mathcal{A}[\![0]\!]\sigma_4] = \{x \mapsto 0, y \mapsto 2\}$
- $\sigma_6 \equiv \sigma_5[y \mapsto \mathcal{A}[\![y + 1]\!]\sigma_5] = \{x \mapsto 0, y \mapsto 3\}$

- $\sigma_7 \equiv \sigma_6[x \mapsto \mathcal{A}[\![x + 1]\!]\sigma_6] = \{x \mapsto 1, y \mapsto 3\}$
- $\sigma_8 \equiv \sigma_7[y \mapsto \mathcal{A}[\![y + 1]\!]\sigma_7] = \{x \mapsto 1, y \mapsto 4\}$
- $\sigma_9 \equiv \sigma_8[x \mapsto \mathcal{A}[\![x + 1]\!]\sigma_8] = \{x \mapsto 2, y \mapsto 4\}$
- $\sigma_{10} \equiv \{x \mapsto 2, y \mapsto 4\}$
- $\sigma_{11} \equiv \sigma_{11}[\text{DV}(\text{var } y := 2;) \mapsto \sigma_2] = \{x \mapsto 2, y \mapsto 5\}$
- $\sigma_{12} \equiv \{x \mapsto 2, y \mapsto 5\}$ (σ_{fin})

2. Semántica SOS

$$\begin{aligned}
\mathcal{T}_1 &= \langle x := 1, \sigma_0 \rangle \Rightarrow \sigma_0[x \mapsto \mathcal{A}[\![1]\!]\sigma_0] \text{ [ass}_{\text{sos}}] \\
\mathcal{T}_2 &= \frac{\mathcal{T}_1}{\langle (x := 1; y := 5), \sigma_0 \rangle \Rightarrow \langle y := 5, \sigma_1 \rangle} \text{ [comp}_{\text{sos}}^2] \\
\mathcal{T}_3 &= \frac{\mathcal{T}_2}{\langle S_1, \sigma_0 \rangle \Rightarrow \langle S_2, \sigma_1 \rangle} \text{ [comp}_{\text{sos}}^1] \\
\mathcal{T}_4 &= \langle y := 5, \sigma_1 \rangle \Rightarrow \sigma_1[y \mapsto \mathcal{A}[\![5]\!]\sigma_1] \text{ [ass}_{\text{sos}}] \\
\mathcal{T}_5 &= \frac{\mathcal{T}_4}{\langle S_3, \sigma_1 \rangle \Rightarrow \langle S_4, \sigma_2 \rangle} \text{ [comp}_{\text{sos}}^2] \\
\mathcal{T}_6 &= \langle \text{var } y := 2; , \sigma_2 \rangle \Rightarrow_D \langle \varepsilon, \sigma_3 \rangle \text{ [var}_{\text{sos}}] \\
\mathcal{T}_7 &= \langle \varepsilon, \sigma_3 \rangle \Rightarrow_D \sigma_3 \text{ [none}_{\text{sos}}] \\
\mathcal{T}_8 &= \frac{\mathcal{T}_6, \mathcal{T}_7}{\langle S_4, \sigma_2 \rangle \Rightarrow \langle S_5, \sigma_3 \rangle} \text{ [begin - block}_{\text{sos}}] \\
\mathcal{T}_9 &= \langle x := x + 5, \sigma_3 \rangle \Rightarrow \sigma_3[x \mapsto \mathcal{A}[\![x + 5]\!]\sigma_3] \text{ [ass}_{\text{sos}}] \\
\mathcal{T}_{10} &= \frac{\mathcal{T}_9}{\langle S_6, \sigma_3 \rangle \Rightarrow \langle \text{for } x := 0 \text{ to } 2 \text{ do } y := y + 1, \sigma_4 \rangle} \text{ [comp}_{\text{sos}}^2] \\
\mathcal{T}_{11} &= \frac{\mathcal{T}_{10}}{\langle S_5, \sigma_3 \rangle \Rightarrow \langle S_7, \sigma_4 \rangle} \text{ [comp}_{\text{sos}}^1] \\
\mathcal{T}_{12} &= \langle \text{for } x := 0 \text{ to } 2 \text{ do } y := y + 1, \sigma_4 \rangle \Rightarrow \langle S_8, \sigma_4 \rangle \text{ [for}_{\text{sos}}] \\
\mathcal{T}_{13} &= \frac{\mathcal{T}_{12}}{\langle S_7, \sigma_4 \rangle \Rightarrow \langle S_9, \sigma_4 \rangle} \text{ [comp}_{\text{sos}}^1] \\
\mathcal{T}_{14} &= \frac{\langle S_8, \sigma_4 \rangle \Rightarrow \langle S_{10}, \sigma_4 \rangle \text{ [if - tt}_{\text{sos}}]}{\langle S_9, \sigma_4 \rangle \Rightarrow \langle S_{11}, \sigma_4 \rangle} \text{ [comp}_{\text{sos}}^1] \\
\mathcal{T}_{15} &= \langle x := 0, \sigma_4 \rangle \Rightarrow \sigma_4[x \mapsto \mathcal{A}[\![0]\!]\sigma_4] \text{ [ass}_{\text{sos}}] \\
\mathcal{T}_{16} &= \frac{\mathcal{T}_{15}}{\langle (x := 0; y := y + 1), \sigma_4 \rangle \Rightarrow \langle y := y + 1, \sigma_5 \rangle} \text{ [comp}_{\text{sos}}^2] \\
\mathcal{T}_{17} &= \frac{\mathcal{T}_{16}}{\langle S_{11}, \sigma_4 \rangle \Rightarrow \langle S_{12}, \sigma_5 \rangle} \text{ [comp}_{\text{sos}}^1] \\
\mathcal{T}_{18} &= \langle y := y + 1, \sigma_5 \rangle \Rightarrow \sigma_5[y \mapsto \mathcal{A}[\![y + 1]\!]\sigma_5] \text{ [ass}_{\text{sos}}]
\end{aligned}$$

$$\begin{aligned}
\mathcal{T}_{19} &= \frac{\mathcal{T}_{18}}{\langle S_{13}, \sigma_5 \rangle \Rightarrow \langle \text{for } x := x + 1 \text{ to } 2 \text{ do } y := y + 1, \sigma_6 \rangle} [\text{comp}_{\text{sos}}^2] \\
\mathcal{T}_{20} &= \frac{\mathcal{T}_{19}}{\langle S_{12}, \sigma_5 \rangle \Rightarrow \langle S_{14}, \sigma_6 \rangle} [\text{comp}_{\text{sos}}^1] \\
\mathcal{T}_{21} &= \langle \text{for } x := x + 1 \text{ to } 2 \text{ do } y := y + 1, \sigma_6 \rangle \Rightarrow \langle S_{15}, \sigma_6 \rangle [\text{for}_{\text{sos}}] \\
\mathcal{T}_{22} &= \frac{\mathcal{T}_{21}}{\langle S_{14}, \sigma_6 \rangle \Rightarrow \langle S_{16}, \sigma_6 \rangle} [\text{comp}_{\text{sos}}^1] \\
\mathcal{T}_{23} &= \frac{\langle S_{15}, \sigma_6 \rangle \Rightarrow \langle S_{17}, \sigma_6 \rangle [\text{if} - \text{tt}_{\text{sos}}]}{\langle S_{16}, \sigma_6 \rangle \Rightarrow \langle S_{18}, \sigma_6 \rangle} [\text{comp}_{\text{sos}}^1] \\
\mathcal{T}_{24} &= \langle x := x + 1, \sigma_6 \rangle \Rightarrow \sigma_6[x \mapsto \mathcal{A}[\![x + 1]\!]\sigma_6] [\text{ass}_{\text{sos}}] \\
\mathcal{T}_{25} &= \frac{\mathcal{T}_{24}}{\langle (x := x + 1; y := y + 1), \sigma_6 \rangle \Rightarrow \langle y := y + 1, \sigma_7 \rangle} [\text{comp}_{\text{sos}}^2] \\
\mathcal{T}_{26} &= \frac{\mathcal{T}_{25}}{\langle S_{18}, \sigma_6 \rangle \Rightarrow \langle S_{12}, \sigma_7 \rangle} [\text{comp}_{\text{sos}}^1] \\
\mathcal{T}_{27} &= \langle y := y + 1, \sigma_7 \rangle \Rightarrow \sigma_7[y \mapsto \mathcal{A}[\![y + 1]\!]\sigma_7] [\text{ass}_{\text{sos}}] \\
\mathcal{T}_{28} &= \frac{\mathcal{T}_{27}}{\langle S_{13}, \sigma_7 \rangle \Rightarrow \langle \text{for } x := x + 1 \text{ to } 2 \text{ do } y := y + 1, \sigma_8 \rangle} [\text{comp}_{\text{sos}}^2] \\
\mathcal{T}_{29} &= \frac{\mathcal{T}_{28}}{\langle S_{12}, \sigma_7 \rangle \Rightarrow \langle S_{14}, \sigma_8 \rangle} [\text{comp}_{\text{sos}}^1] \\
\mathcal{T}_{30} &= \langle \text{for } x := x + 1 \text{ to } 2 \text{ do } y := y + 1, \sigma_8 \rangle \Rightarrow \langle S_{15}, \sigma_8 \rangle [\text{for}_{\text{sos}}] \\
\mathcal{T}_{31} &= \frac{\mathcal{T}_{30}}{\langle S_{14}, \sigma_8 \rangle \Rightarrow \langle S_{16}, \sigma_8 \rangle} [\text{comp}_{\text{sos}}^1] \\
\mathcal{T}_{32} &= \frac{\langle S_{15}, \sigma_8 \rangle \Rightarrow \langle S_{17}, \sigma_8 \rangle [\text{if} - \text{tt}_{\text{sos}}]}{\langle S_{16}, \sigma_9 \rangle \Rightarrow \langle S_{18}, \sigma_8 \rangle} [\text{comp}_{\text{sos}}^1] \\
\mathcal{T}_{33} &= \langle x := x + 1, \sigma_8 \rangle \Rightarrow \sigma_8[x \mapsto \mathcal{A}[\![x + 1]\!]\sigma_8] [\text{ass}_{\text{sos}}] \\
\mathcal{T}_{34} &= \frac{\mathcal{T}_{33}}{\langle (x := x + 1; y := y + 1), \sigma_8 \rangle \Rightarrow \langle y := y + 1, \sigma_{10} \rangle} [\text{comp}_{\text{sos}}^2] \\
\mathcal{T}_{35} &= \frac{\mathcal{T}_{34}}{\langle S_{18}, \sigma_8 \rangle \Rightarrow \langle S_{12}, \sigma_{10} \rangle} [\text{comp}_{\text{sos}}^1] \\
\mathcal{T}_{36} &= \langle y := y + 1, \sigma_{10} \rangle \Rightarrow \sigma_{10}[y \mapsto \mathcal{A}[\![y + 1]\!]\sigma_{10}] [\text{ass}_{\text{sos}}] \\
\mathcal{T}_{37} &= \frac{\mathcal{T}_{36}}{\langle S_{13}, \sigma_{10} \rangle \Rightarrow \langle \text{for } x := x + 1 \text{ to } 2 \text{ do } y := y + 1, \sigma_{11} \rangle} [\text{comp}_{\text{sos}}^2] \\
\mathcal{T}_{38} &= \frac{\mathcal{T}_{37}}{\langle S_{12}, \sigma_{10} \rangle \Rightarrow \langle S_{14}, \sigma_{11} \rangle} [\text{comp}_{\text{sos}}^1] \\
\mathcal{T}_{39} &= \langle \text{for } x := x + 1 \text{ to } 2 \text{ do } y := y + 1, \sigma_{11} \rangle \Rightarrow \langle S_{15}, \sigma_{11} \rangle [\text{for}_{\text{sos}}] \\
\mathcal{T}_{40} &= \frac{\mathcal{T}_{39}}{\langle S_{14}, \sigma_{11} \rangle \Rightarrow \langle S_{16}, \sigma_{11} \rangle} [\text{comp}_{\text{sos}}^1] \\
\mathcal{T}_{41} &= \frac{\langle S_{15}, \sigma_{11} \rangle \Rightarrow \langle \text{skip}, \sigma_{11} \rangle [\text{if} - \text{ff}_{\text{sos}}]}{\langle S_{16}, \sigma_{11} \rangle \Rightarrow \langle S_{19}, \sigma_{11} \rangle} [\text{comp}_{\text{sos}}^1] \\
\mathcal{T}_{42} &= \frac{\langle \text{skip}, \sigma_{11} \rangle \Rightarrow \sigma_{11} [\text{skip}_{\text{sos}}]}{\langle S_{20}, \sigma_{11} \rangle \Rightarrow \langle S_{21}, \sigma_{11} \rangle} [\text{comp}_{\text{sos}}^2] \\
\mathcal{T}_{43} &= \langle S_{21}, \sigma_{11} \rangle \Rightarrow \sigma_{11} [\text{end} - \text{block}_{\text{sos}}]
\end{aligned}$$

Sentencias Abreviadas

$S_1 = (x := 1; y := 5; \textbf{begin var } y := 2; (x := x + 5; \textbf{for } x := 0 \textbf{ to } 2 \textbf{ do } y := y + 1) \textbf{ end}) \quad (\mathbf{S}_{\text{ini}})$
 $S_2 = (y := 5; \textbf{begin var } y := 2; (x := x + 5; \textbf{for } x := 0 \textbf{ to } 2 \textbf{ do } y := y + 1) \textbf{ end})$
 $S_3 = (y := 5; \textbf{begin var } y := 2; (x := x + 5; \textbf{for } x := 0 \textbf{ to } 2 \textbf{ do } y := y + 1) \textbf{ end})$
 $S_4 = \textbf{begin var } y := 2; (x := x + 5; \textbf{for } x := 0 \textbf{ to } 2 \textbf{ do } y := y + 1) \textbf{ end}$
 $S_5 = (x := x + 5; \textbf{for } x := 0 \textbf{ to } 2 \textbf{ do } y := y + 1; \textbf{restore}([x := 1] [y := 5], \textbf{var } y := 2;))$
 $S_6 = (x := x + 5; \textbf{for } x := 0 \textbf{ to } 2 \textbf{ do } y := y + 1)$
 $S_7 = (\textbf{for } x := 0 \textbf{ to } 2 \textbf{ do } y := y + 1; \textbf{restore}([x := 1] [y := 5], \textbf{var } y := 2;))$
 $S_8 = \textbf{if } 0 \leq 2 \textbf{ then } x := 0; y := y + 1; \textbf{for } x := x + 1 \textbf{ to } 2 \textbf{ do } y := y + 1 \textbf{ else skip}$
 $S_9 = (\textbf{if } 0 \leq 2 \textbf{ then } x := 0; y := y + 1; \textbf{for } x := x + 1 \textbf{ to } 2 \textbf{ do } y := y + 1 \textbf{ else skip}; \textbf{restore}([x := 1] [y := 5], \textbf{var } y := 2;))$
 $S_{10} = (x := 0; y := y + 1; \textbf{for } x := x + 1 \textbf{ to } 2 \textbf{ do } y := y + 1)$
 $S_{11} = (x := 0; y := y + 1; \textbf{for } x := x + 1 \textbf{ to } 2 \textbf{ do } y := y + 1; \textbf{restore}([x := 1] [y := 5], \textbf{var } y := 2;))$
 $S_{12} = (y := y + 1; \textbf{for } x := x + 1 \textbf{ to } 2 \textbf{ do } y := y + 1; \textbf{restore}([x := 1] [y := 5], \textbf{var } y := 2;))$
 $S_{13} = (y := y + 1; \textbf{for } x := x + 1 \textbf{ to } 2 \textbf{ do } y := y + 1)$
 $S_{14} = (\textbf{for } x := x + 1 \textbf{ to } 2 \textbf{ do } y := y + 1; \textbf{restore}([x := 1] [y := 5], \textbf{var } y := 2;))$
 $S_{15} = \textbf{if } x + 1 \leq 2 \textbf{ then } x := x + 1; y := y + 1; \textbf{for } x := x + 1 \textbf{ to } 2 \textbf{ do } y := y + 1 \textbf{ else skip}$
 $S_{16} = (\textbf{if } x + 1 \leq 2 \textbf{ then } x := x + 1; y := y + 1; \textbf{for } x := x + 1 \textbf{ to } 2 \textbf{ do } y := y + 1 \textbf{ else skip}; \textbf{restore}([x := 1] [y := 5], \textbf{var } y := 2;))$
 $S_{17} = (x := x + 1; y := y + 1; \textbf{for } x := x + 1 \textbf{ to } 2 \textbf{ do } y := y + 1)$
 $S_{18} = (x := x + 1; y := y + 1; \textbf{for } x := x + 1 \textbf{ to } 2 \textbf{ do } y := y + 1; \textbf{restore}([x := 1] [y := 5], \textbf{var } y := 2;))$
 $S_{19} = (\textbf{skip}; \textbf{restore}([x := 1] [y := 5], \textbf{var } y := 2;))$
 $S_{20} = (\textbf{skip}; \textbf{restore}([x := 1] [y := 5], \textbf{var } y := 2;))$
 $S_{21} = \textbf{restore}([x := 1] [y := 5], \textbf{var } y := 2;) \quad (\mathbf{S}_{\text{fin}})$

Leyenda de Estados

- $\sigma_0 \equiv \emptyset \ (\sigma_{\text{ini}})$
- $\sigma_1 \equiv \sigma_0[x \mapsto \mathcal{A}[1]\sigma_0] = \{x \mapsto 1\}$
- $\sigma_2 \equiv \sigma_1[y \mapsto \mathcal{A}[5]\sigma_1] = \{x \mapsto 1, y \mapsto 5\}$
- $\sigma_3 \equiv \{x \mapsto 1, y \mapsto 2\}$
- $\sigma_4 \equiv \sigma_3[x \mapsto \mathcal{A}[x + 5]\sigma_3] = \{x \mapsto 6, y \mapsto 2\}$
- $\sigma_5 \equiv \sigma_4[x \mapsto \mathcal{A}[0]\sigma_4] = \{x \mapsto 0, y \mapsto 2\}$
- $\sigma_6 \equiv \sigma_5[y \mapsto \mathcal{A}[y + 1]\sigma_5] = \{x \mapsto 0, y \mapsto 3\}$
- $\sigma_7 \equiv \sigma_6[x \mapsto \mathcal{A}[x + 1]\sigma_6] = \{x \mapsto 1, y \mapsto 3\}$
- $\sigma_8 \equiv \sigma_7[y \mapsto \mathcal{A}[y + 1]\sigma_7] = \{x \mapsto 1, y \mapsto 4\}$
- $\sigma_9 \equiv \{x \mapsto 1, y \mapsto 4\}$
- $\sigma_{10} \equiv \sigma_8[x \mapsto \mathcal{A}[x + 1]\sigma_8] = \{x \mapsto 2, y \mapsto 4\}$
- $\sigma_{11} \equiv \sigma_{10}[y \mapsto \mathcal{A}[y + 1]\sigma_{10}] = \{x \mapsto 2, y \mapsto 5\} \ (\sigma_{\text{fin}})$